

in Italy became acknowledged as a success story in the open source space!

Moral of the story: Open source hardware is no longer a dream. It is reality for both the creators and users of technology.

More about FLOSH

Free/libre and open source hardware (FLOSH) extends the concept of open source software to the realm of hardware too—that is, it refers to computers, microcontrollers, electronics and even laptop casings designed in the same way as open source software. The details about the hardware, that is, complete information about how it can be used, the bill of materials, the schematics, the reference design, etc, are made free for all to understand and use. Some believe that a piece of hardware is truly free only if it has also been created using free software, or in a format that can be accessed with free software.

necessary? After all, what could one do with the reference design of a microcontroller? Not everybody can become a manufacturer of hardware, unlike in the case of software, right? It is necessary to understand at this point of time that it is not really a matter of cost or production, but it is also about the ability to understand and use a product better, and adapt it to one's needs if required. More so, today even the cost aspect has been overcome to some extent.

When asked if open source hardware is necessary, D'Souza answers: "In the context of one's rights, my answer is a very emphatic 'yes.' The reason being that there have been in recent times many attempts to lock down hardware (and software) in an attempt to control what one can do within a computing ecosystem. The initial attempts at restriction were trivial in a sense, aimed at gaming machines. But

the earlier constraints of very high costs. In fact, many pieces of 'hardware' inside an average PC (graphics, wireless, media converters, etc) require a piece of software (known as 'firmware') downloaded into them before they work as intended. Changing this piece of firmware will change the functionality of the hardware, by changing the physical interconnections in the device."

It is clear from D'Souza's explanation that the difference between hardware and software has significantly blurred now, making the arguments presented in favour of free software relevant to the hardware space too. D'Souza emphasises that open source hardware is not something esoteric, but it is prevalent and in wide use.

It is for you and me!

Who benefits from FLOSH? Product manufacturers, innovators, hobbyists, students—anybody who is interested in manufacturing such open sourced hardware, using it as part of other products or adapting the boards to suit specific requirements. In short, every consumer.

"FLOSH lowers the entry barrier to innovation. It also brings the strength of massive collaboration—a hallmark of human society and conveniently brushed under the carpet over the last 100 years—to bear upon innovations. By means of these two it makes many solutions to a problem possible, rather than force a one-size-fits-all attitude. Indeed, there are numerous projects not related to the IT field that use the FLOSH paradigm for bringing change in the industry. One such is the Open-car project," says D'Souza.

Syed Mohammed Khasim, lead developer-Open Platforms, Texas Instruments (TI) India, categorically lists the beneficiaries of open source hardware as follows:

Consumers. Open source hardware benefits the consumers as they know what is in the device and can tune it according to their needs rather than going with a closed kit. Today, engineering has become easy and people wish to build their own devices rather

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"Firstly, a clarification: by free we mean freedom, not gratis. Secondly, a piece of hardware consists of many layers of physical and logical devices. Hence the term hardware may also have more than one interpretation, depending on which layer or groups of layers we are referring to," explains J.T. D'Souza, managing director of Sparc Systems (www.sparcsys.com). In today's context, open source hardware most often refers to the 'designs' or abstract representations of hardware, licensed under various free, libre or open sourced licences such as LGPL/BSD-like licences and Creative Commons.

Has the time come?

Articles published (including some in *Linux Today* magazine) a decade ago question the effectiveness of the open source concept in the realm of hardware. Is open source hardware

it is now increasingly being used to prevent usage of media and software. For example, you cannot read a DVD with region encoding if your DVD player is not authorised for that region. The restrictions are in the DVD player, and cannot be circumvented, unless one understands complex hardware. Ironically, DVD players manufactured in China will read everything."

"Further, nowadays it is possible to manufacture general-purpose field-programmable gate array (FPGA) devices very cheaply. These devices are an array of gates, which can be reprogrammed to become a central processing unit (CPU), math processor, digital signal processor (DSP) or any other piece of hardware. The programming is mostly done using hardware definition languages like VHDL. So by using such FPGAs and a program, it is possible to easily build unique hardware without